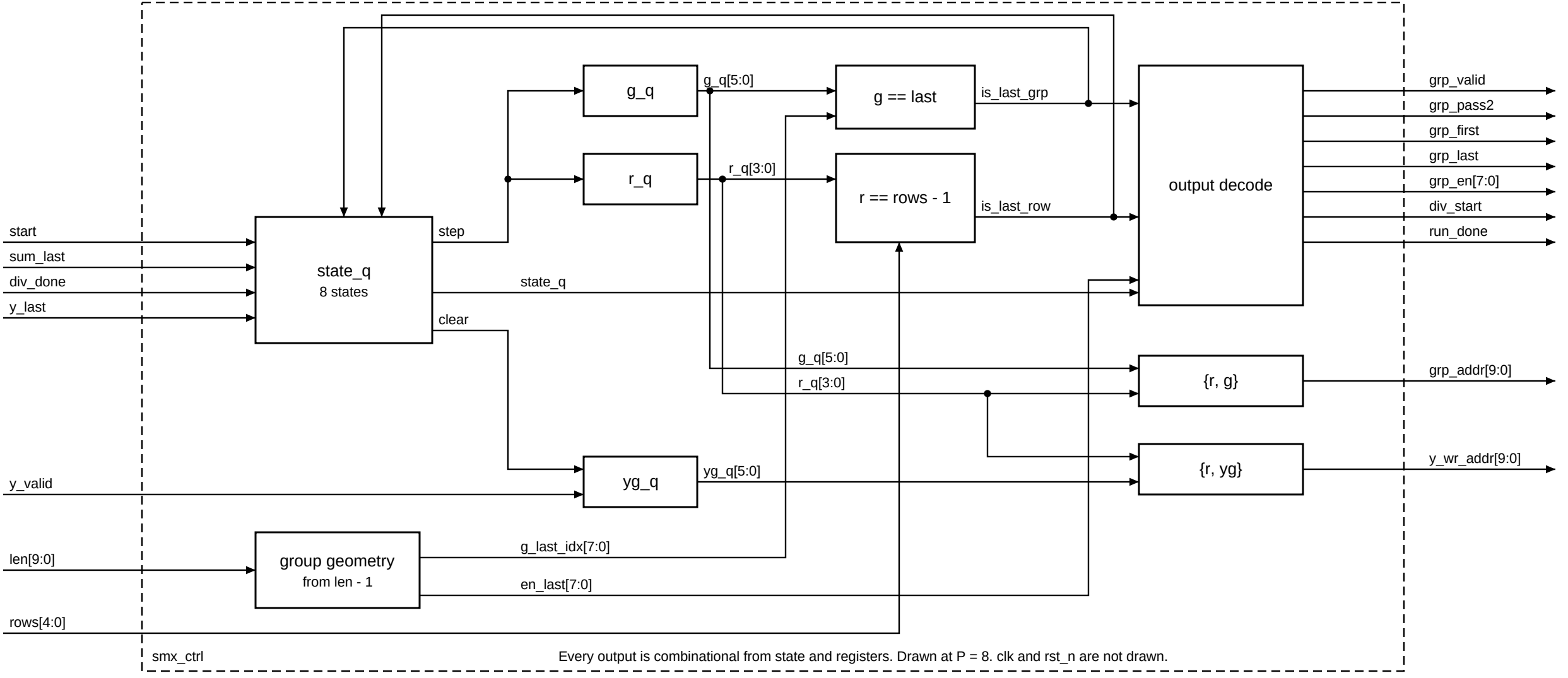
start
sum_last
div_done
y_last
y_valid
len[9:0]
rows[4:0]
state_q
8 states
step
clear
g_q
r_q
yg_q
group geometry
from len - 1
g_q[5:0]
r_q[3:0]
yg_q[5:0]
g_last_idx[7:0]
en_last[7:0]
g == last
r == rows - 1
is_last_grp
is_last_row
state_q
g_q[5:0]
r_q[3:0]
output decode
{r, g}
{r, yg}
grp_valid
grp_pass2
grp_first
grp_last
grp_en[7:0]
div_start
run_done
grp_addr[9:0]
y_wr_addr[9:0]
smx_ctrl
Every output is combinational from state and registers. Drawn at P = 8. clk and rst_n are not drawn.